



Day 10 Summary

Today I learned:

- Arrays
- Loops
- Dictionaries
- Sets
- Extensions
- Collection operations
- Reusable helper methods

◆ Array Operations

Add Value

```
fruits.append("Orange")
```

Remove Value

```
fruits.remove(at: 1)
```

Count Values

```
print(fruits.count)
```

◆ Loops

Used `for-in` loops to iterate through collections.

```
for fruit in fruits {  
    print(fruit)  
}
```

Number Loop:

```
for number in 1...10 {  
    print(number)  
}
```

◆ Dictionaries

Dictionaries store data in **Key** → **Value** pairs.

```
var student = [  
    "name": "Adarsh",  
    "city": "Kanpur"  
]
```

◆ Safe Dictionary Access

Dictionary access returns Optional values because keys may not exist.

```
print(student["name"] ?? "No Name")
```

◆ Sets

Sets store only unique values.

```
var numbers: Set = [1,2,3,4,4,5]

print(numbers)
```

◆ When to Use Sets?

Use Sets when uniqueness matters.

Examples:

- Unique usernames
 - Unique IDs
 - Tags
-

◆ Extensions

Extensions add new functionality to existing types without modifying original code.

```
extension Int {

    func square() -> Int {
        return self * self
    }
}
```

Usage:

```
print(6.square())
```

◆ String Extension Example

```
extension String {

    func welcomeMessage() {
        print("Hello, \(self)")
    }
}
```

```
    }  
}  
  
let name = "Adarsh"  
  
name.welcomeMessage()
```



Important Interview Questions

1 What is an Array in Swift?

An Array stores multiple values of the same type in an ordered collection.

2 How does Array indexing work?

Arrays use indexes starting from 0.

Example:

```
0 → Apple  
1 → Banana  
2 → Cherry
```

3 What is the difference between Array and Set?

Array	Set
Ordered collection	Unordered collection
Allows duplicates	Stores unique values
Uses indexes	No indexes

4 What is a Dictionary in Swift?

A Dictionary stores data in Key → Value pairs.

5 Why does Dictionary access return Optional?

Because the requested key may not exist.

6 What is an Extension in Swift?

An Extension adds new functionality to an existing type without modifying original code.

7 What does `self` mean inside an Extension?

`self` refers to the current object instance.

Example:

```
print("Hello, \(self)")
```

8 Why are Extensions important in iOS development?

Extensions help:

- Organize code
 - Create reusable logic
 - Improve readability
 - Build cleaner architecture
-

Quick Revision

- Array = Ordered collection
- Dictionary = Key → Value pairs
- Set = Unique values only
- `for-in` loop = iteration
- Extension = Add functionality to existing types
- `self` = current object